START FROM SCRATCH

Week 3: Understand, Build, Present

Your readiness brief for a week of RAG learning and a one-day, team-based build sprint.

This week marks a shift.

Move beyond a one-week build cycle. First, build the mental model. Then bring it to life together in a hackathon-style day of focused collaboration.

Your Week 3 rhythm

1

Prepare independently

Complete the assigned RAG articles and videos. Arrive with the concepts, questions, and examples already in mind.

2

Plan with your team

Agree on a practical solution, split responsibilities, choose at least one coding environment, and use GitHub to collaborate.

3

Build together in one day

Use the central meeting room as the headroom. Work from breakout areas as needed, regroup often, and return with a complete solution by end of day.

4

Reserve time to present

Do not leave your story to the final minutes. Prepare the demo and discussion while there is still time to refine the work.

Before the hackathon

Individual readiness

- Complete all pre-reads and videos.
- Have a coding environment ready to use.
- Bring a simple build plan and your questions.

Team readiness

- Align early on the proposed solution.
- Involve your designated lead when needed.
- Use GitHub to share, review, and integrate work.

Presentation standard

15 minutes per team

5-minute live demo + 10-minute discussion

Keep slides minimal. Keep the problem statement short. Talk less, show more.

Be ready to communicate

Leadership | Technical | Sales | Teaching

Explain decisions clearly, use technical language, and adapt the message to your audience.

Aim for work that is useful, tested, and grounded in real needs. Compare your approach with other teams and existing products, then adopt the best practices that fit your solution. Present, receive feedback, and learn from every other team.

WEEK 3 LEARNING PACK

RAG: Learn the concept before you build

Retrieval-Augmented Generation (RAG) helps an AI system retrieve relevant information from a knowledge source and use it to produce a grounded response.

Suggested learning sequence

1. RAG fundamentals

A beginner-friendly walkthrough of the core RAG idea and pipeline.

Open resource

2. RAG crash course

Build a broad mental model before the team starts making technical choices.

Open resource

3. RAG types and architectures

Explore the range of RAG patterns and the trade-offs behind choosing one.

Open resource

4. Retrieval and reranking

Understand how systems find relevant sources, use keyword, vector, or hybrid search, and improve results through reranking.

Open resource

5. Additional RAG video

Use this as an additional perspective during preparation or when the team needs a refresher.

Open resource

What success looks like

By the end of the day, your team should have:

A working solution, a clear explanation of its design, evidence of team collaboration, and a focused presentation ready for a 5-minute demo plus 10-minute discussion.

Bring curiosity, preparation, and a willingness to test your ideas. The goal is not only to ship a prototype - it is to learn how thoughtful teams turn a concept into a credible solution.